healthiar : : CHEAT SHEET

Package overview

Input

Baseline health data
- Mortality
- Morbidity
- …

Optional
`prepare_lifetable()`
Converts muti-year life table, to single year

Exposure
- Population weighted
- Distribution
- …

Optional
`prepare_exposure()`
Converts gridded data to a format usable for attribute_health()

Exposure response function
- Linear
- Log-linear
- …

Other inputs (optional)
- Geo units
- Age groups
- Sex
- …

Calculation

Attribute health impacts

`attribute_health()`
Calculates attributable health impact due to exposure. Can also calculate YLD

Embedded `get_risk()`
Evaluates the exposure response function

Embedded `get_paf()`
Calculates population attributable fraction due to exposure

OR

`attribute_lifetable()`
Calculates years of life lost due to exposure (YLL)

Multiple Impacts

Optional: method to create a second scenario

Optional `attribute_mod()`
Modifies an attribute_health() scenario

`multiexpose()`
Aggregates multiple attribute_health() impacts from different exposures

`compare()`
Compares the impact of two attribute_health() scenarios

Embedded `get_pif()`
Calculates change in impact due to change in exposure scenario

Postprocess Impacts

`daly()`
Calculate disability-adjusted life years (YLL + YLD)

`summarize_uncertainty()`
Calculates uncertainty of impacts with Monte Carlo simulations

`standardize()`
Calculates age standardized health impacts from multiple age groups

Social equality

Optional: socialize() or monetize() without attribute_health()

Optional `prepare_mdi()`
Calculates values for the BEST-COST multidimensional deprivation index

`socialize()`
Calculates impact difference due tow social indicator

Monetization of health impact

Optional `get_inflation_factor()`
Calculates inflation factor based on inflation rate

Optional `get_discount_factor()`
Calculates discount factor based on discount rate

`cba()`
Cost-benefit analysis

`monetize()`

`discount()`
Monetize() without valuation (only %)

Output

attributable **health** impact

health impact attributable to **social** factors

monetized attributable impact

healthiar : : CHEATSHEET

General Tipps

Input format

Format

Compatible: Single value or vector as numerical(s) or string(s)

Incompatible: Lists, data frames, matrices, tibbles
(Exception: "info" argument is compatible with lists, tibbles and data frames)

Structure

For input **tidy structure** is recommended!

bhd	exp_a	exp_b	exp_c	$rr_{central}$	…
15000	17	15	13	1.4	…

OK

geo_Id	exp	$rr_{central}$	bdh	…
"a"	17	1.4	15000	…
"b"	15	1.4	15000	…
"c"	13	1.4	15000	…

BEST- OPTION

Baseline health data (**bhd**), Exposure (**exp**), Exposure response function (**erf**)

Function input (example with attribute_health() function)

Enter input manually

```
results <- attribute_health(
exp_central = c(17,15,13),
bdh_central = c(15000,15000,15000),#or only 1500
erf_shape = "linear",
rr_central = c(1.4,1.4,1.4),#or only 1.4
rr_increment = c(10,10,10),#or only 10
geo_id_micro = c("a", "b", "c"))
```

Enter input from table

```
data <- tidy_table_above
results <- attribute_health(
exp_central = data$exp
bdh_central = data$bhd
erf_shape = "linear",
rr_central = data$relative_risk
rr_increment = data$increment
geo_id_micro = data$geo_id)
```

Output structure

attribute_health() as example

→ Other functions (e.g. **monetize()**, **socialize()**, **compare()**) have similar output structures

Output (as nested list)

```
├─ health_main                   # main results (central uncertainty path)
│    ├ impact                    # attributable health impact
│    ├ pop_fraction              # population attributable fraction (PAF)
│    ├ rr_at_exp                 # riskt at exposure
│    └ …
└─ health_detailed               # detailed and interim results
     ├─ input_args               # status of all possible input arguments
     ├─ input_table              # entered inputs to relevant arguments
     ├─ results_raw              # all results for all uncertainty and group combinations
     ├─ results_by_geo_id_macro  # results aggregated at macro geo unit(s)
     ├─ results_by_geo_id_micro
     ⋮
```

Other possible tibbles: results_by_sex, results_by_age_group, results_by_exp_category, results_by_year…

The output of attribute_health() … is the attributable health impact
… can be used as input for many other functions in heathiar (see first page)

Find more information on …

- **Package Website**
- **GitHub Repository**
- **Vignette**
- **CRAN**

attribute_health()

Pathways – decisions to make

decisions	argument	options
1. Choose risk approach	apporach_risk	RR; AR
2. Choose exposure type	exp_central	Single; Distribution
3. Choose erf	erf_eq_central	Formula; Spline function
	or (only for RR) erf_shape + rr_central + rr_increment	Formula; Spline function; Linear; log-log
Optional extensions		
4. Add cutoff to erf	cutoff_central	Examples: 1
5. Set upper and lower bounds	… _upper	Examples: Value(s) in erf_eq_central +1
	… _lower	Value(s) in erf_eq_central -1
6. Add Years lived with disability(YLD)	dw_central	Examples: 0.5
7. Geographical analysis	geo_id_micro	Paris; Milan; Rome …
	geo_id_macro	France; Italy …
8. Make Groups	sex	Female; Male
	age_group	20-30; 30-40 …

Risk approach

Relative Risk (RR)

exp → erf (evaluate increment) → rr_at_exp ⇨ PAF × bhd = attributable health impact

Required Inputs:

bdh: bdh_central
exp: exp_central
erf: erf_eq_central or erf_shape, rr_central and rr_increment

```
results <- attribute_health(
Apporach_risk = "relative_risk",
exp_central = 8.85,
bdh_central = 30747,
erf_shape = "log_linear",
rr_central = 1.369,
rr_increment = 10)
```

Absolut Risk (AR)

exp erf → absolute_risk_as_percent × #people exposed = attributable health impact

Required Inputs:

bdh: Non
exp: exp_central, pop_exp
erf: erf_eq_central

```
results <- attribute_health(
Apporach_risk = "absolute_risk",
exp_central = 57.5,
pop_exp = 387500,
erf_eq_central = "78.9-3.1*c+0.1*c^2",
cutoff_central= 5)
```

Exposure type

RR or AR

Single exposure

- **One exp** leads to one impact
- If **geo_id:** one exp per geo_id
- Similar for AR

bhd	exp	erf	$rr_{central}$
15000	17	"linear"	1.36

One Impact

RR

Exposure Distribution

- If **Multiple exp** lead to one impact
- If **geo_id:** multiple exp per geo_id

Additionally required: prop_pop_exp

- Sum = 1
- default = 1/#exp (here 4)

→ All inputs except of exp and prop_pop_exp have one fixed value

prop_pop_exp	exp	erf	$rr_{central}$	bdh
0.2	17	"linear"	1.36	15000
0.2	15	"linear"	1.36	15000
0.35	13	"linear"	1.36	15000
0.25	12	"linear"	1.36	15000

One Impact

AR

Exposure Distribution

- **Multiple exp** leads to one impact
- If **geo_id:** one exp per geo_id
- pop_exp behaves similar for AR as prop_pop_exp for RR

→ All inputs except of exp and pop_exp have one fixed value

pop_exp	exp	erf
3000	17	"70-3*c+0.04*c^2"
3000	15	"70-3*c+0.04*c^2"
5250	13	"70-3*c+0.04*c^2"
3750	12	"70-3*c+0.04*c^2"

One Impact

Geographical analysis

Grouping variables: geo_id_micro, geo_id_macro

Grouping effects:

1. Every group needs unambiguous input values
(exception: for exposure distribution exp, pop_exp and prop_pop_exp can be ambiguous)

geo_id_micro	bhd	erf	Prop_pop_exp	$rr_{central}$	bhd
"paris"	20000	"linear"	0.4	1.36	17
"paris"	30000	"linear"	0.6	1.36	15
"rome"	15000	"log_log"	0.3	1.52	15
"rome"	15000	"linear"	0.7	1.52	13
	✗	✗	✓	✓	✓

2. For every group one single Impact is calculated

geo_id_macro	geo_id_micro	bhd	exp	erf	$rr_{central}$
"france"	"paris"	30000	17	"linear"	1.54
"france"	"lyon"	25000	15	"linear"	1.36
"italy"	"rome"	27000	15	"linear"	1.52
"italy"	“milan"	20000	13	"linear"	1.43

results_by_…

geo_id_micro	geo_id_macro
Impact Paris	Impact France
Impact Lyon	
Impact Rome	Impact Italy
Impact Milan	

Sub-group analysis

Grouping variables: age_group, sex, info

1. **Grouping effects** work similar as for geo_id_macro and geo_id_micro (see above)
2. Sub-groups and Geographical analysis can be combined

Example: c("paris", "paris", "rome", "rome") and c("male", "female", "male", "female") = c("paris.male", "paris.female", "rome.male", "rome.female") = four groups